Selected-Frame Longitudinal Organization in UNESCO Cultural Heritage Sites: Within-Representation Diagnostics for a 3° Harmonic Frame

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Abstract

The full-corpus Rayleigh test for 3° spectral periodicity in UNESCO Cultural and Mixed World Heritage longitudes ($N = 1011$) is null ($p = 0.3716$), and a global anchor-discovery sweep is marginal ($p = 0.0753$); against these boundary conditions, this paper reports a selected-frame count excess under a fixed 3° harmonic-deviation representation referenced to a Levantine ∼35°E corridor, for which Mount Gerizim (35.269°E) is one historically motivated candidate point rather than a uniquely identified anchor. The target claim is representational, not invariant: the chosen frame yields one primary Tier-A+ excess in the UNESCO corpus, characterized through several diagnostic angles, with a related 3° pattern in two Eurasian corpora that share the corpus's geography.

Within-frame characterization. Inside the fixed frame, the full corpus shows Tier-A+ harmonic-node enrichment (1.31×, $p_{\text{adj}} = 0.0465$). This count excess is further characterized by: (i) a Spearman-style cluster-asymmetry pattern in which harmonic bands containing a precisely aligned site host 4.24× more sites than empty bands ($p_{\text{adj}} = 0.0012$); (ii) a phase-rotation null in which the selected phase ranks at the 100.0$^{\text{th}}$ percentile of all 300 envelope phases; (iii) Tier-A++ enrichment in the domed/spherical sub-population (3.06×, $p_{\text{adj}} = 0.0416$), with leave-3-out worst-case robustness but with Null A consistent with the pattern being a consequence of dome-site geography; (iv) a unit sweep in which 3° is the dominant spacing and adjacent non-harmonic units (1.8°, 2.1°, 2.7°, 3.3°, 3.6°) are null, while 4° and 8° reduce to nodes shared with the 3° grid; and (v) the same 3° frame producing related structure in the Wikidata Q180987 stupa corpus and the OWTRAD attested Silk Road network. Across the full audit ($k = 44$), 6 results survive Bonferroni correction.

Scope. The dome A++ result is sharper than its longitude footprint under a restricted-draw null (Null C, marginal across three windows) but is consistent with within-dome geographic concentration under Null A; it is therefore reported as a sub-population characterization within the selected frame, not as independent evidence of coordinated placement. The Wikidata and OWTRAD analyses are not geography-independent replications: UNESCO cultural sites, Wikidata stupas, and

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OWTRAD route nodes are all concentrated in the Eurasian corridor. Their shared 3° agreement is nevertheless the most interesting empirical feature of the analysis and warrants further investigation. The priority next step is application of the frozen pipeline by an independent team to heritage corpora outside the Eurasian corridor. Pre-modern sub-degree continental longitude determination is not a documented capability; we propose no historical mechanism, and we read the selected-frame organization as a hypothesis-generating regularity whose historical status is open. The analytical choices were made in contact with the data, and the pipeline, data, and reproduction scripts are openly deposited (Zenodo `10.5281/zenodo.19938283`).

Keywords: selected-frame analysis; longitudinal organization; UNESCO World Heritage; spatial statistics; harmonic deviation; sensitivity analysis

1 Introduction

The full-corpus Rayleigh statistic for 3° periodicity in raw UNESCO longitudes is null ($p = 0.3716$), and the global anchor-discovery sweep is marginal ($p = 0.0753$). This study therefore does not begin from an invariant periodicity claim. It reports a selected-frame analysis of the externally curated UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$) under a fixed 3° harmonic-deviation representation referenced to the Levantine ∼35°E corridor, for which Mount Gerizim (35.269°E; see §2.2) is one historically motivated candidate point rather than a uniquely identified anchor. The contribution is representational rather than invariant: the chosen frame yields one primary Tier-A+ count excess in the UNESCO corpus, characterized through several diagnostic angles, with a related 3° pattern in two Eurasian heritage corpora that share the corpus's geography.

The frame was selected in contact with the data (3° spacing, Gerizim anchor, dome keyword set, von Mises tier widths) and then held fixed across the analyses reported here; the open question is whether this representation organizes the data in a way that random permutations of the same representation do not, not whether the data exhibit invariant 3° periodicity in raw longitude.

Boundary conditions vs. within-frame diagnostics. We distinguish two classes of test, and we treat them as having different evidentiary roles. Phase-invariant tests (the full-corpus Rayleigh statistic for 3° periodicity, the global anchor-discovery sweep) ask whether the corpus exhibits a global 3° periodicity that survives marginalization over phase and anchor. These tests are negative or marginal here ($p = 0.3716$ and $p = 0.0753$, respectively; §4.1), and we report them as *boundary conditions*: they delimit what we do not claim. We do not claim invariant global periodicity, and we do not claim that the Levantine corridor is statistically privileged in a phase-free sense. Conditional, within-frame diagnostics ask whether, given the fixed selected representation, the corpus is unusually organized relative to permutation and bootstrap nulls drawn from the same representation. These diagnostics include Tier-A+ harmonic-node enrichment, harmonic-band cluster asymmetry, the phase-rotation null evaluated within the selected 3° circle, the unit-sweep contrast against adjacent non-harmonic spacings, the dome/stupa sub-population analysis, leave-n-out and multi-scale threshold checks, and portability checks in the Wikidata Q180987 and OWTRAD corpora. Several of these views are mathematically related rather than independent confirmations: the cluster-asymmetry test partly characterizes how the A+ count is distributed across harmonic bands, the phase-rotation percentile locates the selected phase within the same count envelope, and the dome A++ result must be read with the Null A caveat that dome-site geography can account for much of the pattern. The positive result is therefore a count excess under the chosen frame and its behavior under these characterizations, not a set of independent proofs of the same effect.

This is a different kind of claim from invariant periodicity. The selected frame is anchor-specific and unit-specific by construction, and we treat that specificity as part of what is being characterized, not as something to be argued away. Whether the within-frame organization reflects historical intentionality, geographic and cultural channeling along Eurasian corridors, inscription selection effects, or some combination of these is not decided by the present analysis; we discuss candidate explanations in §7.1 and §7.4 without endorsing any.

Site-level structure under the frame. Sites within ≤ 16.6 km of a 3°-grid node are designated Tier-A+ ($P_0 = 10\%$); the full corpus yields 132 A+ sites against 101.1 expected. The excess is sharpened in domed monuments (§4.3), although Null A is consistent with that sub-population pattern being a consequence of where dome sites already are, and it is concentrated in harmonic bands containing a precisely aligned site (§4.5; 4.24× denser). A global anchor sweep shows that the within-frame enrichment is stable across the ∼35°E corridor (§5.3). Tier boundaries are set from the angular dispersion of an independently curated but geographically related stupa corpus (Wikidata Q180987, $N = 229$; $\hat{\sigma} = 0.1922°$), fixing the $0.25\hat{\sigma}/0.50\hat{\sigma}/1.00\hat{\sigma}$ tier widths without reference to UNESCO data. Because UNESCO cultural sites, Wikidata stupas, and OWTRAD route nodes are all concentrated in the Eurasian corridor, cross-corpus agreement at 3° is reported as an empirical observation for further testing rather than as geography-independent replication.

Methodological safeguards. Prior longitude-alignment studies were criticised for investigator-assembled site lists (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). This study differs: (1) the UNESCO corpus is externally curated and fixed; (2) the 3° period was identified through iterative examination of the corpus and subsequently characterized by a unit sweep (§6.2)—these are not equivalent to a parameter-free discovery, and we do not present them as such; (3) analytical choices are fixed at the level of the deposited codebase for reanalysis, and all results can be regenerated from the public specification (Tenenbaum, 2026).

Exploratory status. Because all analytical choices were informed by iterative examination of the UNESCO corpus, this study reports an exploratory empirical regularity rather than a hypothesis-confirming test. The relevant scientific questions are: (1) Is the within-frame organization internally consistent under sensitivity checks? (2) Was the reference phase chosen on independent grounds, and is anchor-specificity itself a coherent feature of the selected representation? (3) What additional evidence would constitute external validation? These questions are addressed in §6.1 and §6.2; we return to the third in the Conclusion.

2 Background

2.1 Prior Work on Spatial Periodicity and Sacred Landscapes

Statistical evaluation of spatial periodicity in monument distributions has a long and contested history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Magli (Magli, 2013, 2016) has applied Monte Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties.

2.2 Reference Phase and Candidate Anchors

The harmonic deviations in this study are computed from Mount Gerizim (35.269°E) as the *a priori* reference phase. Gerizim is chosen because of its independently documented *tabbur ha-aretz* ("navel of the earth") cosmographic role in Samaritan tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), a justification that predates the analysis entirely.

Caveat on anchor identification. The statistical analysis cannot uniquely identify which physical location, if any, historically defined the grid origin. The von Mises mixture model (§3.3) recovers a phase offset relative to any fixed reference frame, but the fitted phase is an estimate subject to sampling uncertainty. Any UNESCO site within the A+ tier ($\lesssim$16.6 km) of the Gerizim-anchored grid could serve as an alternative reference point and reconstruct an essentially identical periodicity. The primary finding is therefore the existence of the 3° periodic structure within the ~35°E corridor; Mount Gerizim is presented as a historically motivated candidate anchor based on independent textual evidence (*tabbur ha-aretz* cosmographic tradition; Purvis 1968), not as a conclusion drawn from the statistical analysis. A site-as-anchor ranking audit (Supplementary Audit S6) is consistent with this: the top-ranked competing anchors, including the Silk Roads: Zarafshan-Karakum Corridor (62.21°E, deviation 0.061° from Gerizim harmonic $n = 9$), the Birthplace of Jesus (35.21°E), and Old City of Sana'a (44.21°E), all lie within the A+ tier of the Gerizim grid and are therefore co-grid points rather than independent alternatives.

A global anchor sweep (§5.3) shows the result is not sensitive to the exact coordinate: every longitude in the x.25°E band produces the same enrichment, and both Gerizim and Jerusalem (35.232°E, ≈4.1 km south) place the corridor in the global top 2.3% of 36,000 trial anchors. Mecca ($p \approx 0.2825$, ns), Mt. Meru ($p \approx 0.3956$, ns), and Mt. Kailash ($p \approx 0.3956$, ns) yield no significant enrichment, consistent with phase specificity, even if the phase origin cannot be uniquely attributed.

Gerizim's coordinate is from the UNESCO tentative list (ref. 5706, DMS E35 16 08, State of Palestine, 2012; `https://whc.unesco.org/en/tentativelists/5706/`) and is not inscribed. Jerusalem is an inscribed corpus member, self-excluded when used as anchor.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset (UNESCO World Heritage Centre, 2024) contains 1248 inscribed nominations. UNESCO designates each site *Cultural* (criteria i–vi), *Natural* (criteria vii–x), or *Mixed*; sites with cultural heritage alongside natural values receive Mixed and are included here. Retaining Cultural and Mixed (−235 Natural-only) and excluding 2 entries lacking geocoordinates (both linear route networks) yields $N = 1011$ sites.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E (see §2.2). Gerizim is not inscribed and does not appear in the standard corpus. Jerusalem is inscribed and scored as a regular site in Tests 1–4 (A+ at 4.1 km under Gerizim). For sensitivity sweeps (§5.3), Gerizim is added as a synthetic entry so each anchor self-excludes symmetrically.

For a site at longitude λ, the deviation from the nearest 3° harmonic is $\delta = |\lambda - 35.269| \bmod 3°$ (folded to $[0, 1.5°]$). The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 16.6$ km for Tier-A+) are equatorial approximations computed as $\delta \times 111$ km/°, becoming smaller at higher latitudes; the angular criterion is consequently conservative at high latitudes. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.15°/1.5° = 10\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 101.1 ± 9.6 expected A+ ($\sim$10%), indicating the geometric null is not materially inflated by longitude clumping.

3.3 Tier Classification

Tier boundaries are derived from the cross-corpus von Mises dispersion scale $\hat{\sigma} = 0.1922°$ (stupa corpus, $\hat{\kappa} = 6.17$; see "Von Mises cross-corpus characterisation" below). The three A-tiers correspond to $0.25\hat{\sigma}$, $0.50\hat{\sigma}$, and $1.00\hat{\sigma}$ respectively.

Tier A++ $0.25\hat{\sigma}$; $\delta \leq 0.06°$ ($\lesssim 6.66$ km); $P_0 = 4\%$.

Tier A+ $0.50\hat{\sigma}$ (primary threshold); $\delta \leq 0.15°$ ($\lesssim 16.6$ km); $P_0 = 10\%$.

Tier A $1.00\hat{\sigma}$; $\delta \leq 0.3°$ ($\lesssim 33$ km); $P_0 = 20\%$.

Tier B Middle zone, not near any harmonic or inter-harmonic midpoint.

Tier C/C−/C−− Symmetric mirror of A/A+/A++ measured from the nearest inter-harmonic midpoint.

Von Mises cross-corpus characterisation. We fitted $f(\theta \mid \kappa) \propto e^{\kappa \cos\theta}$ (von Mises + uniform mixture) to the harmonic phase deviations of two external corpora: OWTRAD Silk Road nodes ($N = 1674$; $\hat{\kappa} = 9.58$) and Wikidata Q180987 stupas ($N = 229$; $\hat{\kappa} = 6.17$). The stupa corpus yields $\hat{\sigma} = \kappa^{-1/2} = 0.1922°$; $0.25\hat{\sigma} = 0.048°$ (A++), $0.50\hat{\sigma} = 0.096°$ (A+), and $1.00\hat{\sigma} = 0.1923°$ (A) match the tier thresholds to within rounding (Table 1). The N-weighted pooled $\hat{\sigma}$ is 0.1588°. Both external corpora are concentrated in the Eurasian corridor, so their use here fixes a cross-corpus angular scale within the same geographic domain rather than providing geography-independent validation.

Table 1: Tier boundaries as multiples of the cross-corpus von Mises dispersion $\hat{\sigma} = 0.1922°$ ($\hat{\kappa} = 6.17$, Wikidata Q180987 stupa corpus, $N = 229$).

$\hat{\sigma}$ multiple	Threshold	P_0
$0.25\hat{\sigma} = 0.048°$	0.06° (≈ 6.66 km)	4%
$0.50\hat{\sigma} = 0.096°$	0.15° (≈ 16.6 km)	10%
$1.00\hat{\sigma} = 0.1923°$	0.3° (≈ 33 km)	20%

3.4 Test Populations

Keyword populations are approximate. The raw sweep tolerates known false positives while the context-validated variant (Test 2x) applies sentence-level gating for greater precision. Both populations are reported.

Test 1 (full corpus, primary). Full corpus ($N = 1011$), no keyword filter; binomial test of whether the global A+ rate exceeds the 10% geometric null.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being non-architectural false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter; tests whether A+ harmonics host more total sites than non-A+ harmonics, independent of any morphological keyword.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus based on year of inscription into the UNESCO corpus, no keyword filter.

Test 5 (founding classifier, post-hoc). A post-hoc keyword classifier identifies site types beyond domed monuments that fall on harmonic nodes; it carries no weight in the primary evidential chain (Appendix A.2).

3.5 Multiple-Comparisons Protocol

Full reported family ($k = 44$). The multiple-comparisons headline treats the 44 tests in the FDR audit as the correction family. Six results survive Bonferroni at $k = 44$: dome/spherical A++ (binomial $p_{\text{adj}} = 0.0416$, with the Null A geographic caveat discussed in §4.3), full-corpus A+ ($p_{\text{adj}} = 0.0465$), cluster permutation ($p_{\text{adj}} =< 0.0001$), harmonic Mann-Whitney ($p_{\text{adj}} = 0.0012$), sequential-peak permutation ($p_{\text{adj}} = 0.0279$), and evolution combined A++ (binomial $p_{\text{adj}} = 0.0208$).

Non-primary reduced correction ($k = 3$, Tests 1–3 only).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 10% null.
- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 10% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

The $k = 3$ family was not pre-specified and is reported only as a sensitivity analysis. The primary multiple-comparisons report is the full $k = 44$ audit above.

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is descriptive context; a Mantel-Haenszel stratified test and logistic regression with region as a covariate assess the gradient robustly (§4.6); it is not used as a Bonferroni-survivor claim. Test 2b is a sensitivity variant of Test 2. Test 5 (founding classifier) is post-hoc (Appendix A.2). World religion sub-group analyses (§4.2) use Fisher exact tests and are not Bonferroni-corrected. Tests A, C, D, and E (§4.5) assess the 3° periodicity structure: Test A (full-corpus Rayleigh) is a negative control structurally insensitive to the sparse A+ sub-population; Tests C and D are conditional anchor-specific tests; Test E is the anchor-discovery correction. Test B is tautological by construction and excluded from both tables.

3.6 Simulation Null Models

Three simulation approaches (100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5.

3.7 Computational Reproducibility

All statistics are generated by an automated pipeline; every script writes to a shared results store and LaTeX macros are regenerated without manual editing. Scripts, audit files, and reproduction instructions are catalogued in Supplementary S1; see also §A.2).

4 Results

4.1 Boundary Conditions: Invariant Tests Delimit What We Do Not Claim

Three phase-invariant checks delimit the scope of the present claim before the within-frame diagnostics are reported. They tell us what the result is *not*: it is not an invariant global 3° periodicity in raw longitude, and it is not a phase ranking that survives a full anchor-discovery correction. We treat these results as boundary conditions on the interpretation, not as a verdict on the within-frame organization characterized in §4.2–§4.5.

(i) The full-corpus Rayleigh test returns null ($p_{\text{perm}} = 0.3716$ ns). The raw longitude distribution does not exhibit smooth sinusoidal periodicity at 3°. Lattice-node proximity (the present target) and spectral periodicity (what Rayleigh tests) are distinct mathematical objects; a clumpy, anchor-specific concentration of mass near a small subset of harmonic nodes can coexist with a null Rayleigh statistic, and we interpret the conjunction in those terms rather than as a contradiction.

(ii) A phase-envelope scan over all origins $\phi \in [0, 3°)$ (300 phases at 0.01° resolution) records strong phase localization within the selected 3° circle: full-corpus A+ counts span 83–132 (median 100.0), and Gerizim's phase (2.2690° mod 3°) sits at the 100.0^{th} percentile (132 A+), the 96.7^{th} for domes (18/90; range 4–21). Within the selected representation, the chosen phase is unusually good; this is a within-frame diagnostic, not an invariant claim.

(iii) A degree-denominated Monte Carlo unit sweep identifies 3° as the dominant period (§6.2); adjacent non-harmonic units are null (§5.1). The unit sweep does not require an anchor and is reported here as the unit-side complement to the phase scan: together they indicate that 3° is the spacing at which the within-frame count excess concentrates and that the selected phase is the phase at which the concentration is sharpest, while the anchor-discovery correction across all anchors is marginal (Test E, $p = 0.0753$, †; §4.5).

The remainder of the Results characterizes the corpus *conditional on* this representation: harmonic-node enrichment in the full corpus (§4.2), morphological sub-population sharpening (§4.3), and cluster-asymmetry behavior at the harmonic level (§4.5). These sections characterize the selected frame's count excess and its

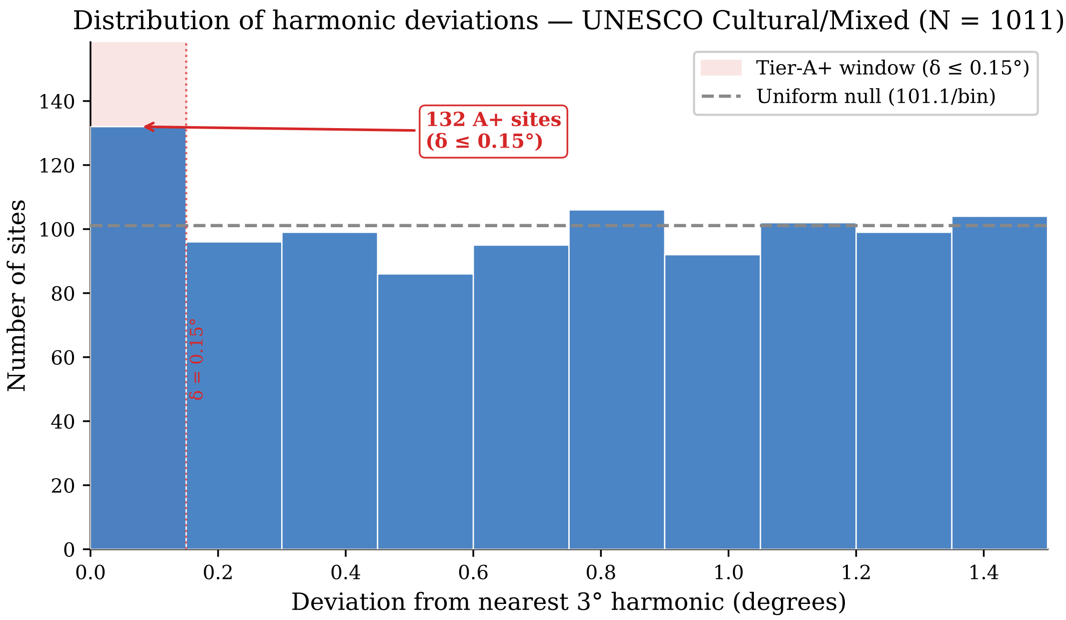

Figure 1: Distribution of harmonic deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.15°$, shaded) above the uniform null (dashed line) is consistent with the corpus-level signal against which Tests 2 and 3 characterize morphological and spatial specificity.

distributional consequences; they are not presented as phase- or anchor-marginalized confirmation.

4.2 Global Enrichment (Test 1, Primary)

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 13.1% (Figure 1) (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), above the 10% geometric null, which assumes uniform longitude distribution even though the corpus does not satisfy that assumption; empirical-null results are reported alongside (binomial $p = 0.0011$; Bonferroni $p_{\text{adj}} = 0.005$, $k = 3$). The observed enrichment is 1.31×, modest relative to the dome subpopulation (2.00×) and the harmonic cluster asymmetry. For transparency, the non-primary reduced correction for Tests 1–3 ($k = 3$) is reported in Table 4; the primary multiple-comparisons report is the full $k = 44$ audit in §5.7.

Region-stratified robustness. A region-specific empirical null (Fisher combination of per-region binomials) yields $p = 0.7978$ (ns); the pre-2000 cohort remains significant against the geometric null ($p = 0.0132$ *; §4.6). Both results are included in the FDR audit (Table 4).

World religion sites

Religion sub-populations are defined by keyword sets documented in the supplementary audit archive (Tenenbaum, 2026). A chi-squared test finds no significant A+ rate difference across the five main religion groups ($\chi^2 = 4.753$, df = 4, $p = 0.3136$, ns). Both Buddhism (20/71 A-tier, 28.2%; 19/71 C-band, 26.8%; $p = 0.0052$ **) and Hinduism (11/39 A-tier, 28.2%; 10/39 C-band, 25.6%; $p = 0.0324$ *) show a joint A-tier/C-band signal (multivariate hypergeometric); both persist in tradition-exclusive sites ($p = 0.0123$ *, $p = 0.0862$ †).

4.3 Domed and Spherical Monuments (Test 2, Primary)

The keyword filter is described in §3.4 ($N = 90$).

Tier-A++ concentration (key result). 11 of 90 dome sites (12.2%, 3.06× the 4% null) fall within Tier-A++ ($\delta \leq 0.06°$, $\lesssim 6.66$ km). This concentration is significant across three test statistics: Fisher OR = 2.71, $p = 0.0077$ **; permutation $Z = 3.21$, $p = 0.004$ **; bootstrap $p = 0.006$ **. Dome A++ survives full Bonferroni correction at $k = 44$ (§5.7), but Null A is consistent with the pattern being a consequence of dome-site geography (§4.3).

Tier-A+ enrichment (reduced-correction sensitivity). 18/90 sites at Tier-A+ (20.0% [12.3, 29.8], 2.00×; Fisher OR = 1.77, $p = 0.0346$ *; binomial $p = 0.0032$ **; Bonferroni $p_{\text{adj}} = 0.002$, $k = 3$). Tier-A: 29/90 (32.2%, 1.61×; Fisher $p = 0.0175$ *; binomial $p = 0.0043$ **). The χ^2-uniform test ($p = 0.0732$, †) is consistent with concentration at harmonic nodes. Tier-B (middle zone): 45/90 sites; mean $\delta = 0.6586°$.

Tier-A+ sites: Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out binomial check indicates no single site drives the result: removing any A+ site yields $N = 89$, A+= 17, $p = 0.0067$. Leave-2-out (worst case) across all $\binom{18}{2}$ combinations: $p = 0.0133$. Leave-3-out (worst case): $p = 0.0253$; the evolution corpus (Test 2b) (§4.4) and the A++ concentration are more robust. The A++ leave-3-out worst case across all $\binom{11}{3} = 165$ triples yields $p = 0.0233$, remaining significant at $\alpha = 0.05$ (leave-2-out worst case $p = 0.0087$).

Context-validated robustness (Tests 2x). Sentence-level context validation reduces the dome corpus to $N = 83$ (7 false positives rejected; Appendix A.1). Test 2x Fisher exact (dome sub-pop vs. full corpus, $N = 1011$): A++ (11/83, 3.00×, $p = 0.0041$ **); A+ (16/83, 1.67×, $p = 0.0614$ †); A (26/83, 1.64×, $p = 0.0346$ *). This context validation improves keyword precision but does not remove the Null A caveat that dome-site geography may account for the sub-population pattern.

Keyword-stratified sensitivity (leave-stupa-out). Partitioning the dome population by triggering keyword shows that stupa/stupas ($n = 18$) has the strongest individual signal (A++ OR = 6.94, $p = 0.0057$ **), but dome/tholos/spherical alone ($n = 65$, no stupa keyword) is significant in its own stratum at A++ (OR = 2.93, $p = 0.0193$ *). Mound keywords (tumulus/barrow/kofun) show no enrichment ($n = 0$), consistent with the filter being morphologically discriminating. These keyword-stratified checks are subject to the same Null A geographic caveat as the full dome result.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 71% fall in 20°–120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (100,000 trials).

Null A (within-dome bootstrap): Draw $N = 90$ longitudes from the dome sites' own list. Mean = 18.00 A+, $p = 0.5429$ (ns). This null is consistent with the observed A+ count being a consequence of where dome sites already are: the dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all full-corpus sites within $\pm W°$ of any dome site and bootstrap the expected A+ count. Dome sites (18 A+, 20.0%) nominally exceed the broader footprint at all three windows tested: $\pm 2°$ ($p = 0.0257$, *); $\pm 5°$ ($p = 0.0393$, *); $\pm 10°$ ($p = 0.0442$, *). These marginal results ($p \approx 0.04$ across the three windows) do not survive correction across the window sweep.

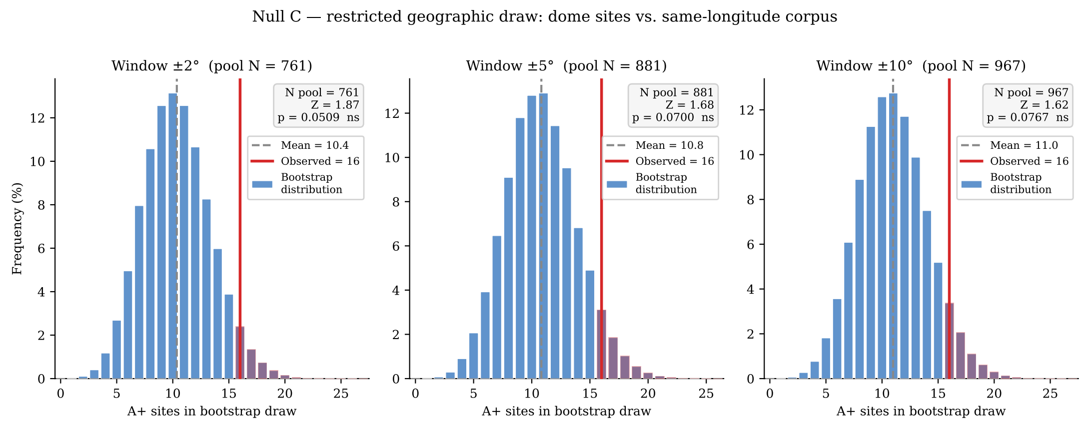

Figure 2: Null C bootstrap distributions for the dome sub-population ($N = 90$, observed A+ $= 18$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 90$ from that pool define the expected A+ distribution (blue bars). The observed count (red line) nominally exceeds the pool mean by $Z \approx 2.7$ in all three windows ($\pm 2^\circ$: $p = 0.0257$, *; $\pm 5^\circ$: $p = 0.0393$, *; $\pm 10^\circ$: $p = 0.0442$, *), before correction across the three tested windows.

Table 2: Test 2b enrichment by evolutionary stage ($N = 110$). Null rates: A++ $P_0 = 4\%$, A+ $P_0 = 10\%$, A $P_0 = 20\%$. Fisher exact tests: stage vs. full corpus background.

		Tier A++ ($\leq$ 6.66 km)		Tier A+ ($\leq$ 16.6 km)		Tier A ($\leq$ 33 km)	
Stage	N	A++	Rate OR (p)	A+	Rate OR (p)	A	Rate OR (p)
Mound[a]	24	2	8.3% 1.57× (0.3880 ns)	2	8.3% 0.60× (0.8430 ns)	6	25.0% 1.15× (0.4653 ns)
Stupa[b]	18	4	22.2% 5.17× (0.0144 *)	6	33.3% 3.44× (0.0216 *)	7	38.9% 2.22× (0.0874 †)
Dome[c]	75	8	10.7% 2.21× (0.0485 *)	13	17.3% 1.44× (0.1663 ns)	23	30.7% 1.58× (0.0576 †)
Combined	**110**	**13**	**11.8% 2.67× (0.0658 **)**	**20**	**18.2% 1.57× (0.0658 †)**	**35**	**31.8% 1.71× (0.0113 *)**

[a]tumulus/barrow/kofun. [b]stupa/stupas. [c]dome/tholos/spherical. OR = odds ratio; p = one-sided Fisher exact vs. full UNESCO cultural corpus ($N = 1011$). Stage N values sum to more than the combined total because 4 sites match multiple stages.

Reconciliation. Null A is consistent with the pattern being a consequence of where dome sites already are, while Null C suggests possible residual morphological specificity within the same longitude footprint. Because the Null C window results are marginal and do not survive correction across the three windows, geographic concentration accounts for much but potentially not all of the observed pattern; this remains an open question.

In the global anchor sweep (§5.3), Gerizim scores 132 A+ (99.7$^{\text{th}}$ percentile); Jerusalem scores 128 (97.7$^{\text{th}}$).

4.4 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extending Test 2 with earthen mound-stage keywords (see §3.4); the combined population ($N = 110$) adds 20 mound-stage sites to the core domed/spherical population, with 4 sites matching both keyword sets.

Stage-by-stage breakdown (Table 2):

The combined population ($N = 110$, A+ $= 18.2\%$; Table 2) is consistent with Test 2. The mound stage does not reach significance (8.3%, OR $= 0.60\times$, Fisher $p = 0.8430$,

Table 3: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.30°$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.30°$ lon)	6.56	4.96	1.32×	
Mann-Whitney U (one-tailed)				0.0025 **
Permutation (10,000 shuffles)			$Z = 3.96$	0.0000 ***
Harmonic-level density (all Tier $\leq$ B sites per 3° harmonic):				
Harmonics with ≥ 1 A+ site (36)	mean = 25.0 sites			
Harmonics with 0 A+ sites (19)	mean = 5.9 sites		4.24×	< 0.0001 ***

ns). The signal is carried by the stupa (A+ 33.3%, OR = 3.44×, $p = 0.0216$ *) and dome sub-stages (A+ 17.3%, OR = 1.44×, $p = 0.1663$ ns). Leave-one-out checks indicate stability. Leave-3-out (worst case, 1140 triples): $p = 0.0371$.

Context-validated robustness (2bx). The context-validated 2bx sample remains robust: ($N = 104$, 6 rejected): A++ (13/104, 2.94×, $p = 0.0021$ **); A+ (18/104, 1.55×, $p = 0.0772$ †); A (32/104, 1.61×, $p = 0.0239$ *).

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5879$ (ns). Null C (restricted $\pm 5°$ draw): $p = 0.0556$ (†).

4.5 Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter (see §3.4). For each site, count other UNESCO sites within $\pm 0.30°$ longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.
A+-bearing harmonics host 4.24× more sites on average than non-A+ harmonics ($p < 0.0001$; Table 3). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.25 vs. 6.49 neighbors; 1.43×, $p = 0.0527$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations) finds the conditional fraction of clustering-matched permutations producing the canonical-unit peak (0.1667) is identical to the unconditional rate (0.1786): the clustering and unit-sensitivity signals are statistically independent.

Rayleigh and anchor-specific checks. The A+-only Rayleigh statistic is omitted from the results because it is a mathematical consequence of the A+ classification criterion (§6.1), not an independent test. The following formal tests address distinct aspects of the periodicity:

Test A (full-corpus Rayleigh periodogram — boundary condition). The full-corpus Rayleigh $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$, ns) at $T = 3°$ is null. We report this as a boundary condition, not a failure of the primary claim. The Rayleigh periodogram asks whether raw longitudes show a phase-marginalized sinusoidal periodicity across the corpus; it is structurally insensitive to sparse, anchor-specific clustering near a subset of fixed lattice nodes, because sites near nodes and anti-nodes are both treated as phase information about the same period. The present

Table 4: Non-primary multiple-comparisons sensitivity summary. Reduced correction for Tests 1–3 only ($k = 3$, $\alpha = 0.05/3$); the primary report is the full $k = 44$ audit.

Test	N	Raw p	p_{adj}, k=3	Status
Test 1 (full corpus)	1011	0.0011^a	0.005	Non-primary sensitivity
Test 2 (dome A+)	90	0.0346^a (OR = 1.77)	0.002	Non-primary sensitivity
Test 3 (cluster asym.)	1011	0.0000^b	0.142	Non-primary sensitivity
Test 4 (temporal)	1011	0.4184^b	—	Descriptive
Test 2b (mound evol.)	110	0.0658^a (OR = 1.57)	N/A	Sensitivity

[a]Fisher exact p vs. rest of corpus (one-sided, enrichment); [b]permutation p. Binomial baselines: Test 2 $p = 0.0032$ **, Test 2b $p = 0.0062$ **.

analysis therefore treats the Rayleigh null as evidence against invariant 3° periodicity, while leaving the selected-frame node-proximity count excess to be evaluated by the conditional tests below.

Test C (circular-shift anchor sensitivity). Displacing the anchor by a uniform random offset in $[0°, 360°)$ and recounting A+ sites yields a null mean of 101.05 vs. the observed 132 ($Z = 2.88$, $p_{\text{perm}} = 0.0061$, **). A positive anchor-specific test complementary to Test D.

Test D (targeted von Mises mixture, full corpus). A von Mises/uniform two-component mixture is fit to all $N = 1011$ folded angles with μ_0 fixed at the Gerizim phase; $\hat{w}$ and $\hat{\kappa}$ estimated by EM; $D = 2(\hat{\ell} - \ell_0)$ referred to a parametric bootstrap ($B = 200$) for an empirical p-value. On the full corpus: $D = 8.6103$, $\hat{w} = 0.0312$ (i.e. ≈3.1% of all sites), $\hat{\kappa} = 36.70$ (angular SD ≈ 0.0788°), $p_{\text{boot}} = 0.0199$ (*). On the A+ subset alone: $D = 562.7225$, $\hat{w} = 1.0000$, $\hat{\kappa} = 31.89$ (angular SD ≈ 0.0845°), $p_{\text{boot}} = 0.0010$ (***). Tests C and D are consistent with anchor-specific 3° structure rather than a statistical artifact.

Test E (anchor discovery correction). A global sweep of 36,000 trial anchors asks whether the Levantine corridor ranks unusually among all possible anchors ($p = 0.0753$, †). This is a conservative upper bound on the anchor-selection penalty; the strongest pattern rests on Tests 2 and 3, which do not depend on the anchor choice (§6.1).

4.6 Temporal Gradient (Test 4, Descriptive)

The Tier-A++ rate in the founding canon (1978–1984) is 8.0% (11/138; 1.38× the 4% null, $p = 0.0232$ *), declining monotonically toward the null in later cohorts (2000–2025: 4.5%). Cochran-Armitage three-cohort trend test (A++, decreasing): $Z = -1.664$, $p = 0.0481$ (*). The five-cohort trend is marginal ($p = 0.0489$ *). By contrast, the A+ rate shows no significant decreasing trend ($p = 0.4184$ ns), indicating the gradient is specific to the highest-precision tier. The leading non-intentional explanation is a founding-canon selection effect: UNESCO's earliest inscriptions prioritized famous monuments; famous monuments cluster in well-known cultural centers; and those centers are disproportionately located along the Eurasian corridor in which the selected 3° frame is being characterized. The Mantel-Haenszel stratified test is underpowered at the sub-group level; Logistic regression with region as a covariate is consistent with the A++ gradient ($Z = -1.760$, $p = 0.0784$ †, OR = 0.722). The declining rate may also partly reflect UNESCO's expansion into regions where hemispherical traditions are less prevalent. The gradient is treated as descriptive context only (Figure 3).

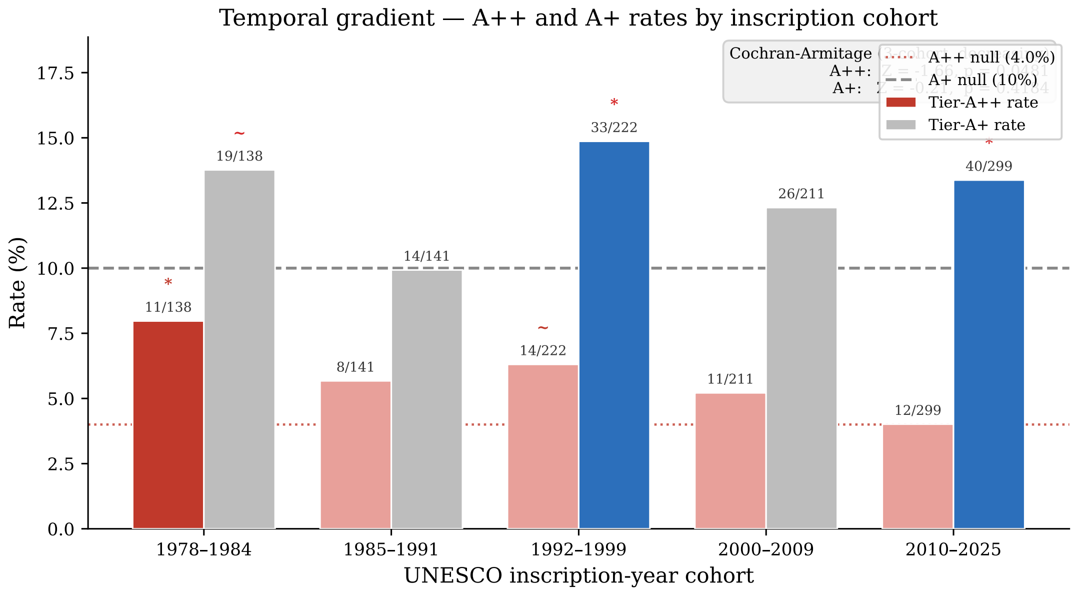

Figure 3: Tier-A++ and Tier-A+ rates by UNESCO inscription-year cohort. The founding canon (1978–1984) shows an elevated A++ rate (8.0%, $p = 0.0232$ *) that declines monotonically across later cohorts toward the 4% null (dotted line; Cochran-Armitage $Z = -1.664$, $p = 0.0481$ *). The A+ rate (right bars) shows no significant trend ($p = 0.4184$ ns).

4.7 FDR audit and Bonferroni correction

The primary multiple-comparisons headline is the full reported family: $k = 44$ tests in the FDR audit.

A comprehensive FDR audit (44 tests; Benjamini & Hochberg 1995) retains 22 at $q < 0.05$, of which 6 survive full Bonferroni at $k = 44$. The 6 full-k survivors are: dome/spherical A++ (binomial $p_{\text{adj}} = 0.0416$, with Null A consistent with dome-site geography accounting for the pattern); full-corpus A+ (binomial $p_{\text{adj}} = 0.0465$); cluster permutation ($p_{\text{adj}} =< 0.0001$); harmonic Mann-Whitney ($p_{\text{adj}} = 0.0012$); sequential-peak permutation ($p_{\text{adj}} = 0.0279$; §4.6); and evolution combined A++ (binomial $p_{\text{adj}} = 0.0208$).

Non-primary reduced correction ($k = 3$, Tests 1–3 only)

Table 4 reports the reduced correction for Tests 1–3 as a sensitivity analysis. The $k = 3$ family was not pre-specified; the full $k = 44$ audit is the primary multiple-comparisons report.

5 Robustness Checks

5.1 Unit Sensitivity Sweep

Varying harmonic spacing from 1.5° to 6° (physical threshold fixed at 16.6 km), the 3° spacing produces the strongest joint significance across both the dome and full-corpus populations. The 1.5° and 6° spacings also reach significance, but both are mathematical consequences of the 3° structure: every 3° harmonic is also a 1.5° harmonic, and 6° harmonics are a strict subset, so neither constitutes an independent signal. Adjacent non-harmonic spacings (1.8°, 2.1°, 2.7°, 3.3°, 3.6°) are non-significant in either population, the key discriminating observation. The 2.4° spacing reaches nominal dome significance ($p = 0.0506$) but not full-corpus ($p = 0.2470$, ns), reflecting unsystematic local overlap with the 3° grid rather than a genuine secondary periodicity. Specifically, 7/16 (43.8%) of 2.4° dome hits align to the 3° grid, indicating the significance is due to overlap.

A degree-denominated Monte Carlo sweep (§6.2) is consistent with this pattern. Candidate intervals at 4° and 8° reach nominal dome significance, but inspection shows that 6 of 7 and all 4 of their hits respectively fall on 12° and 24° nodes shared with the 3° grid, indicating they are consequences of the primary periodicity rather than independent angular units.

5.2 Sensitivity Slope at the Canonical Unit

A fine sweep within ±1% of 3° shows a ±0.3% shift collapses joint significance in at least one population. A geographic bootstrap of the full cultural corpus finds no permutation in 10000 trials that reproduces the canonical-unit sharp peak at either the A^{+} or A^{++} precision tier ($p = 0.0000$ and $p = 1.0000$ respectively, both ***), consistent with 3° as a specific periodicity. The dome-filter specificity test yields marginal results across all tiers (A^{++}: $p = 0.1728$, ns; A^{+}: $p = 0.0999$, ∼; A: $p = 0.0427$, *), consistent with the test being underpowered at $n = 83$ sites. Restricting to the 22 morphologically unambiguous spherical forms (stupa, tholos) recovers significance at the A tier ($p = 0.0239$, *). Tests C and D (§4.5) are consistent with 3° structure.

A log-scale threshold sweep (73 values, 27/28 significant within ±1 log-decade of A+) yields dome-population data-optimal bands A^{++}≈5.0 km, A^{+}≈9.2 km, A≈17.0 km (geometric ratio r^{*}=1.87); the analytically derived tier boundaries (0.06°, 0.15°, 0.3°) are the nearest $\hat{\sigma}$-multiple boundaries to these optima.

5.3 Global Anchor Sweep

A global sweep (0–360°E, 0.01° resolution, 36,000 trial anchors) assesses whether the ∼35°E corridor is unusual globally. The sweep uses the extended corpus described in §3.2 ($N_{\text{ext}} = 1012$), each anchor excluding its own entry, so the focal anchors test against $N = 1011$.

Both anchors place the corridor in the global top 2.3% (36,000 trial anchors, 0–360°E; Gerizim: 132 A+, 99.7$^{\text{th}}$ percentile; Jerusalem: 128 A+, 97.7$^{\text{th}}$ percentile, and the result is anchor-invariant within the 4.1 km corridor window. No other historical meridian reaches significance.

The $x.25$°E periodic structure

Every $x.25$°E longitude (120 anchors) produces identically the maximum A+ count, spaced 3° apart. The label denotes phase ≈ 2.25° mod 3°; Gerizim's phase (2.269°, gap 0.019° ≈ 1.8 km) is within coordinate uncertainty.

Formal statistical tests for the $x.25$° periodicity

The full-corpus Rayleigh ($p_{\text{perm}} = 0.3716$, ns) does not show global periodicity; Test B is tautological. Tests C ($p_{\text{perm}} = 0.0061$, **) and D ($p_{\text{boot}} = 0.0199$, *) are positive anchor-specific tests; Test E ($p = 0.0753$, †, marginal) is the anchor discovery correction. The unit sensitivity sweep (§5.1) and Monte Carlo unit sweep (§6.2) are consistent with 3° as the dominant period without assuming any particular unit.

A dome-phase uniformity check does not distinguish alignment from geographic concentration. Gerizim scores at the 96.5$^{\text{th}}$ percentile among dome sites; all 18 dome A+ sites lie within ±0.15° of the x.25° optimal phase.

Corridor and landmark ranking. Of the 13 A+ sites unique to Gerizim and 9 unique to Jerusalem, 119 are shared; the enrichment belongs to the corridor, not to either anchor individually. Using each UNESCO site as anchor, Gerizim ranks #8 of 1012 (top 2.8%).

5.4 Spatial Autocorrelation

A+ harmonics host 4.24× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$). DEFF correction via ICC (±0.5°, $\rho = 0.187$): DEFF = 1.842, $N_{\text{eff}} = 549$, corrected $p = 0.011$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI [10.50%, 15.43%] excludes the 10% null. Within-group permutation using UNESCO-region × longitude bins reproduced the observed A+ count exactly in every trial, showing this test cannot discriminate alignment from geographic concentration; the Null A and Null C simulations (§4.3) address that question directly.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.060$ †; 1978-1984 canon ($N = 138$) $p = 0.433$ ns; pre-2000 ($N = 501$) $p = 0.490$ ns. (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.070$ †. (3) KDE-smoothed null: expected 101.1 ± 9.6, observed 132, $Z = 3.23$, $p = 0.001$ **. The KDE null ($p = 0.001$ **) is consistent with dome enrichment; permutation and bootstrap are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.004$ **, bootstrap $p = 0.006$ **), with Null A consistent with the A++ pattern being a consequence of dome-site geography. The geographic-concentration nulls (Null A, Null C; §4.3) show that geographic concentration accounts for much but potentially not all of the observed pattern; this remains an open question.

5.6 Pipeline Portability: Wikidata Q180987 Stupa Corpus

The pipeline was applied to $N = 229$ Wikidata-attested stupas (class Q180987) as a within-method portability check using an independently curated, but geographically related, corpus (Tenenbaum, 2026). Because the Wikidata stupa corpus is also concentrated in the Eurasian corridor, this analysis is not a geography-independent replication of the UNESCO result. The Kedu Plain (Java) hosts the densest cluster at the 75° harmonic (75°E from Gerizim), anchored by Borobudur (c. 800 CE, 7.23 km, Tier A+) (Miksic, 1990); the World Peace Pagoda at Lumbini achieves A++ precision (∼555 m) uniquely from Gerizim.

Using the identical anchor and null, the corpus yields 70/229 Tier-A sites (30.6%, 1.53× null; binomial $p < 0.001$, ***; permutation $Z = 1.59$, $p_{\text{perm}} = 0.065$, ∼). No Tier-A+ enrichment is found (binomial $p = 0.021$, *), consistent with cluster-level rather than individual-site precision. Harmonic nodes with at least one Tier-A site carry 2.56× more stupa sites on average than empty nodes, a related harmonic-band density pattern under the same representation.

Geographic tier breakdown and bimodal structure. All regional p-values below are one-sided Fisher exact tests. A sub-regional breakdown reveals a bimodal pattern consistent with the Buddhist and Hindu bimodal enrichment in the UNESCO religion audit (§4.2). The Java/Sumatra node (105°–115°E, near the 75° harmonic at 110.3°E) concentrates 43 of 83 sites at Tier-A (51.8%, OR = 4.74×, $p < 0.001$ ***), rising to 42/54 (77.8%, OR = 18.38×, $p < 0.001$ ***) in the 107°–112°E window, where A+ is also elevated at 31.5% (OR = 4.57×, $p =< 0.001$ ***). The physical heartland, India/Nepal (70°–90°E) and Myanmar/Cambodia (90°–105°E), is A-tier depleted (combined OR = 0.21×, $p < 0.001$ ***, one-sided depletion test) but strongly enriched in the inter-harmonic C-band: Myanmar/Cambodia reaches 35.8% (OR = 4.47×, $p < 0.001$ ***) and the combined heartland 29.2% (OR = 6.00×, $p < 0.001$ ***). This A/C anti-correlation mirrors the bimodal pattern in the religion audit (§4.2). Non-heartland Wikidata stupas (outside 70°–110°E, $N = 102$) show the same A-enriched, C-depleted profile: 50.0% reach Tier-A (OR = 5.68×, $p < 0.001$ ***) with no C-band enrichment (OR = 0.19×, $p = 1.000$ ns).

5.7 Multiple Comparisons

See §3.5 for the full protocol. Gerizim ranks at the 99.7$^{\text{th}}$ percentile; a ±0.3% unit shift collapses joint significance. FDR audit (44 tests) retains 22 at $q < 0.05$; 6 survive Bonferroni at $k = 44$.

6 Limitations

6.1 Circularity: Full Canonical Treatment

Exploratory status and inferential limitations. Because all analytical choices (3° spacing, Gerizim anchor, dome keyword set, tier thresholds) were informed by iterative examination of the UNESCO corpus, this study reports an exploratory finding rather than a hypothesis-confirming test. The relevant scientific questions are: (1) Is the within-frame organization internally consistent under sensitivity checks? (2) Was the reference phase chosen on independent grounds, and is the anchor-specificity itself a coherent feature of the selected representation rather than an artifact of investigator-degrees-of-freedom? (3) What additional evidence would constitute external validation? The first two are addressed below; the third is taken up in the Conclusion.

Anchor-specificity as a property of the selected representation. The within-frame organization characterized in §4.2–§4.5 is by construction anchor-specific: it is a property of the chosen 3°/Gerizim representation, not a claim about a phase that survives marginalization over all anchors. The global anchor sweep (§5.3) is consistent with the enrichment being a property of the ∼35°E corridor: every x.25°E anchor produces the same A+ count, and both Gerizim and Jerusalem place the corridor in the global top 2.3% of 36,000 trial anchors. Gerizim is distinguished within that corridor by pre-existing documentary evidence—the *tabbur ha-aretz* cosmographic role (§2.2)—predating this analysis. We do not present this as a unique solution to an inverse problem; we present Gerizim as one historically motivated point in a corridor whose width is itself part of the result.

The 3° structure is recoverable by any unanchored 120-phase scan within the corridor. Dome enrichment (Test 2) is identical across all 120 x.25°E anchors. Cluster asymmetry (Test 3) requires no keyword filter, but it is still defined by the same A+ classification and therefore characterizes the selected representation rather than supplying an anchor-independent test. The anchor-discovery correction (Test E, $p = 0.0753$, †) is marginal, which we report as a boundary condition (§4.1) rather than a verdict; the strongest within-frame diagnostics (Tests 2 and 3) do not depend on the anchor choice.

6.2 Geographic and Cultural Selection Bias

The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor of Cultural/Mixed sites. The regional distribution is as follows: (Europe & North America: 50%; Arab States: 9%; Asia-Pacific: 23%; Latin Americas: 11%; Africa: 7%). Domed monuments nominally exceed the A+ rate of other monuments from the same longitude footprint (Null C, $p = 0.0393$, *), while Null A ($p = 0.5429$, ns) is consistent with the pattern being a consequence of where dome sites already are.

A Monte Carlo unit sweep (10000 permutations, 18 candidate spacing values from 0.5° to 15°) identifies which angular interval produces clustering beyond geographic concentration without assuming any particular unit. The dominant peak is at 3° ($p < 0.001$), with harmonics at 6° and 12° also significant. Collapsing harmonics, the dome subset yields 1 independent dominant spacings ($P = 0.2248$ under the geographic null), consistent with the selected-frame dome count excess concentrating at 3° rather than at adjacent candidate spacings.

6.3 Americas Negative Control

The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 9.7%, below the 10% null (one-sided depletion: $p = 0.5149$ ns). The sample is too small to treat this as strong depletion, but it does not contradict the primary result.

6.4 Other Limitations

Geocoding precision. UNESCO XML coordinates are administrative centroids: unsystematic noise, no directional bias.

Metrological evidence. A post-hoc numerical correspondence with Babylonian sub-units was noticed and discarded as non-load-bearing (Hunger & Steele, 2019; Robson, 2008; Powell, 1987); all thresholds are degree-denominated from cross-corpus angular dispersion.

7 Discussion

7.1 Alternative Explanations

(1) Founding-canon and inscription selection. UNESCO's founding canon prioritized famous monuments; famous monuments cluster in well-known cultural centers; and well-known cultural centers cluster along the Eurasian corridor. This selection pathway is a leading non-intentional explanation for the elevated early A++ rate reported in §4.6.

(2) Geographic concentration. Null A is consistent with the dome pattern being a consequence of where dome sites already are, while Null C finds that other monuments from the same longitude footprint are nominally less enriched ($p = 0.0393$, *). The Monte Carlo unit sweep (§6.2) is consistent with 3° without assuming any unit.

(3) Morphological recurrence. Hemispherical structures recur across cultures (Aveni, 2001, 2003), but recurrence by itself predicts no longitude specificity.

(4) Random coincidence. The KDE-smoothed null is consistent with dome enrichment ($p = 0.001$ **); permutation ($p = 0.060$ †) and bootstrap ($p = 0.070$ †) are marginal at the A+ level but highly significant for the A++ sub-population (permutation $p = 0.004$ **; bootstrap $p = 0.006$ **), subject to the Null A caveat above. Tests C and D (§4.5) are consistent with anchor specificity and periodicity structure.

7.2 Interpreting Modest Effect Sizes

The observed enrichment (1.31× full corpus, 2.00× domes at A+, 3.06× domes at A++, with the Null A caveat that dome-site geography can account for much of the sub-population pattern) is statistically robust but modest in magnitude. We interpret this as a pattern consistent with partial adherence, selection, or preferential tendency rather than strict observance of a 3° placement rule. Several explanations are consistent with this pattern:

(1) Founding-canon selection effects. UNESCO's earliest inscriptions emphasized famous monuments. Famous monuments are disproportionately located in well-known cultural centers, and those centers are concentrated along the Eurasian corridor. This mechanism does not require intentional longitude placement and is given the same evidentiary weight here as logistical or diffusion explanations.

(2) Logistical convenience. Trade routes and way-stations naturally cluster at intervals determined by travel distance: one day's journey (∼30 km at 30°N latitude) corresponds to roughly 0.3°, so a 3° grid represents about ten such intervals and could capture higher-order staging posts without coordinated placement.

(3) Cultural diffusion with local variation. Buddhist architectural traditions spread along the Silk Roads with transmission hubs at specific longitudes (Lumbini, the Borobudur node at ≈110°E), while local adaptation introduces scatter that flattens the effect size when averaged across the full corpus.

(4) Selective preservation beyond the founding canon. Later inscription and preservation processes may continue to favor culturally significant sites at trade junctions or political centers that happen to align with the corridor, even as the World Heritage List expands geographically.

These explanations remain speculative. The contribution of this paper is to document the empirical regularity and its statistical robustness; determining the historical mechanism, if any, requires independent lines of evidence beyond spatial statistics alone.

7.3 Framework Portability: OWTRAD and Wikidata Corpora

The Wikidata Q180987 stupa corpus and the OWTRAD attested Silk Road network are not intended as geography-independent replications of the UNESCO result. They are targeted out-of-sample tests of the hypothesized Eurasian corridor signal: if the UNESCO pattern were merely an idiosyncrasy of UNESCO inscription practice—a selection-effect artifact of how that particular committee chooses and documents sites—the same phase and enrichment structure would not necessarily appear in independently assembled stupa and route-node datasets that were compiled for entirely different purposes. Their agreement therefore supports corpus-level portability

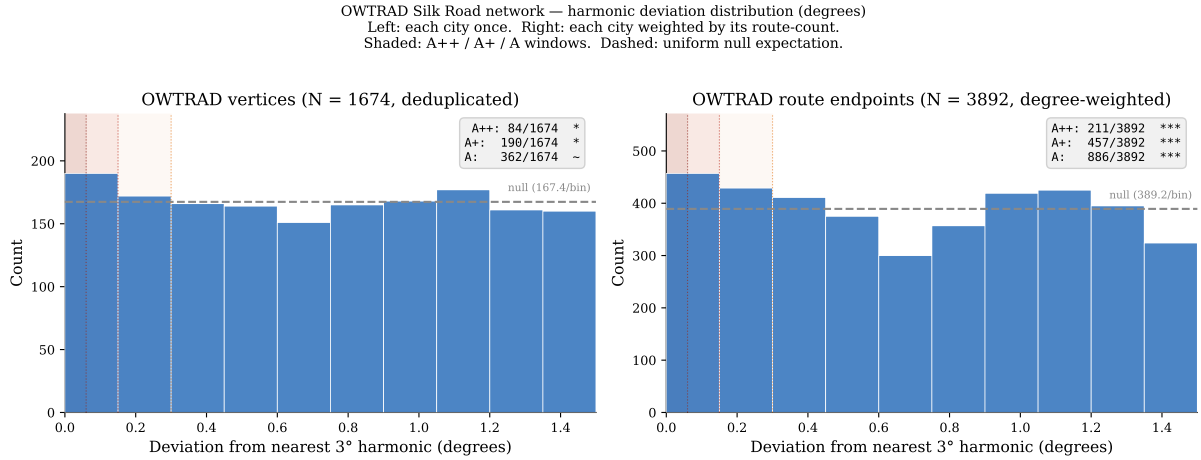

Figure 4: OWTRAD Silk Road endpoint tier distributions: 1674 deduplicated endpoints (left) and 3892 degree-weighted positions (right). Both show significant A++ enrichment (deduplicated: $z = 2.13$, $p = 0.0224$ *; degree-weighted: $z = 4.53$, $p =< 0.0001$ ***).

within the hypothesized Silk Road/stupa domain. It does not constitute geography-independent confirmation, because all three datasets are concentrated in the Eurasian corridor; the agreement is consistent with, but does not require, an explanation that goes beyond shared corridor geography. We report these analyses as portability checks that use the same anchor and parameters as the primary analysis—not as independent confirmation—and we treat the shared 3° pattern as a real empirical observation that is not reducible to the UNESCO corpus alone. The priority for future work is geography-independent validation: application of this framework, with all parameters frozen at the deposited values, by a different research team to heritage corpora from regions outside the Eurasian corridor assembled without reference to this analysis. First, the OWTRAD attested-route network (Ciolek, 2004) (1946 route segments, 1674 unique endpoints). Testing all 3892 endpoint positions across the full edge list (each city weighted by the number of segments it connects), 211 fall at A++ harmonics, 457 at A+, and 886 at A (1.36× null rate, $z = 4.53$, $p =< 0.0001$ *** for A++; $p = 0.0002$ *** for A+; $p =< 0.0001$ *** for A; one-tailed binomial). Testing each unique endpoint once gives 84/1674 at A++, 190 at A+, and 362 at A (1.25×, $z = 2.13$, $p = 0.0224$ * for A++; $p = 0.0377$ * for A+; $p = 0.0524$ † for A; one-tailed binomial). A phase-shift test (random offset of the 3° grid, 100000 iterations) gives $p = 0.0202$ *, consistent with enrichment specific to the Gerizim anchor. A cluster asymmetry test mirrors Test 3 of the primary analysis: harmonics containing at least one A++ endpoint (30 harmonics) carry 2.70× more total route-connectivity than harmonics without one (11 harmonics; $p = 0.0007$ ***, Mann-Whitney one-tailed). The Silk Road corridor is illustrated in Figure 5; harmonic tier distributions for both OWTRAD populations are shown in Figure 4.

7.4 The Mechanism Gap

A literal reading of the within-frame organization—that builders across Eurasia placed monuments with sub-degree longitudinal precision relative to a common reference—would require pre-modern continental longitude determination at a precision that is not a documented historical capability. Babylonian astronomical methods support angular measurement and eclipse timing (Hunger & Steele, 2019; Neugebauer, 1975; Robson, 2008) but do not extend to terrestrial longitude across continental scales,

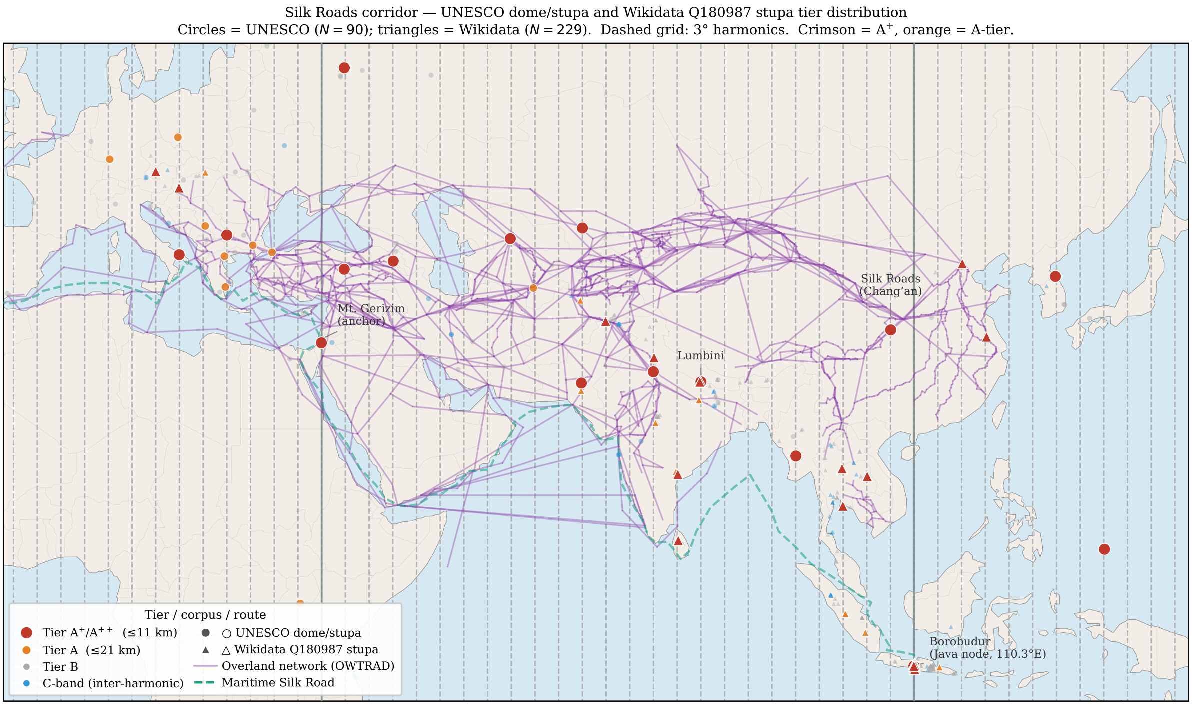

Figure 5: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor. Overland route edges from the OWTRAD attested-route database (Ciolek, 2004). Maritime route is a schematic coastal polyline after Casson (1989), Sen (2014), and Tomber (2008). An interactive version is provided as supplementary material (`supplementary/interactive_map/silkroad_interactive.html`).

and no surveying instrument calibrated in ancient metrological sub-units relevant to such a procedure has been recovered. We take this gap seriously as a constraint on interpretation rather than as a peripheral caveat: any historical reading of the reported regularity must favor explanations that do not require coordinated placement.

This is why we frame the empirical contribution as representation-induced organization rather than as evidence of metrology or coordinated design. The selected 3°/Gerizim/dome representation interacts with the actual geographic and cultural distribution of UNESCO sites—trade-route spacing at characteristic travel intervals, geographic channeling by river systems and passes, the concentration of inscribable cultural heritage along the Eurasian corridor, and inscription selection effects within UNESCO itself—to produce a set of within-frame characterizations. That the selected-frame count excess is sharper than chance under permutation and bootstrap nulls drawn from the same representation is the result we report. Whether the pattern ultimately reflects historical intentionality, the geometry of Eurasian corridor geography under a particular representational choice, or some combination of these, is not decided here, and we explicitly do not propose a mechanism.

8 Conclusion

We report a selected-frame count excess in the longitudinal distribution of the UNESCO Cultural and Mixed corpus under a fixed 3° harmonic-deviation representation referenced to the Levantine corridor. The contribution is representational rather than invariant. Phase-invariant tests—the full-corpus Rayleigh statistic ($p = 0.3716$) and the global anchor-discovery correction ($p = 0.0753$)—are negative or marginal and are reported as boundary conditions: we do not claim invariant global periodicity or a phase-free privileged status for the corridor.

Inside the fixed selected representation, the primary positive result is full-corpus A+ harmonic-node enrichment (1.31×). That count excess is characterized by a Spearman-style harmonic-band asymmetry in which A+-bearing bands host 4.24× more sites than empty bands; phase localization within the selected 3° circle (100.0$^{\text{th}}$ percentile); A++ enrichment in the domed/spherical sub-population (3.06×) with leave-3-out worst-case robustness, while Null A remains consistent with the dome result being a consequence of dome-site geography; a unit sweep in which 3° is the dominant spacing and adjacent non-harmonic units are null while 4° and 8° reduce to nodes shared with the 3° grid; and a related 3° pattern in the Wikidata Q180987 stupa corpus and the OWTRAD attested Silk Road network, both of which share the same broad Eurasian corridor geography. Under full Bonferroni correction across $k = 44$ reported tests, 6 results survive.

The selected 3°/Gerizim representation is anchor-specific and unit-specific by construction; we treat that specificity as part of what is being characterized. Within the corridor, every x.25°E anchor produces the same enrichment, and Mount Gerizim is presented as a historically documented candidate within that corridor (*tabbur ha-aretz*; Purvis 1968), not as a unique solution to an inverse problem. The Wikidata and OWTRAD analyses are framework-portability checks within the Eurasian corridor, not independent geographic confirmation.

What is and is not claimed. We claim that the chosen representation organizes the UNESCO corpus and two related Eurasian corpora in a way that random permutations of the same representation do not. We do not treat mathematically related diagnostics as independent confirmation, and we do not claim invariant 3° periodicity, a unique historical anchor, or a metrological mechanism. We explicitly flag the absence of any documented pre-modern capability for sub-degree continental longitude determination (§7.4) as a constraint on historical interpretation.

External validation and next steps. Because the corpora analyzed here were fixed before this study and selected for cultural importance rather than for spatial analysis, traditional pre-registration is not available as a validation pathway. The shared 3° agreement across UNESCO, Wikidata Q180987, and OWTRAD is the most interesting feature of the analysis, but it does not establish the effect because all three corpora are Eurasian-corridor concentrated. Meaningful external validation would require: (1) application of this framework—with 3° spacing, Gerizim anchor, and dome keyword set fixed as in the Zenodo v1.6.0 deposit—to heritage corpora outside the Eurasian corridor, including national registers not consulted during analysis; (2) adversarial reanalysis by an independent team using the open codebase; or (3) archaeological evidence bearing on whether builders at the identified sites had any access to longitude information at the implied precision. The highest-priority next step is application of the frozen pipeline to non-Eurasian heritage corpora by an independent team. We make all data, code, and methods freely available (§A.2; Zenodo DOI `10.5281/zenodo.19938283`) to facilitate such external checks. The scientific value of this work ultimately depends on whether the selected-frame pattern is found in datasets assembled independently by other researchers outside the corridor studied here.

A Keyword Audit Summary

Full site-by-site audit files are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`; Tenenbaum 2026). Regenerate all with `bash tools/update_all_audits.sh`.

A.1 Dome/Spherical Monument Filter (Tests 2, 2b, 2x, 2bx)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All 90 matching sites are included in Test 2 without context gating.

For Test 2x, ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural context. Unambiguous keywords (*stupa/stupas*, *tholos*) are accepted without validation. The seven rejected sites are: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres.

Test 2b adds earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun* unambiguous; *mound* context-validated) to trace the hemispherical tradition from burial mound through stupa to architectural dome. Test 2bx applies the same architectural context gates to the extended population. Statistics for all four variants are reported in §4.3 and §4.4.

A.2 Founding-Origin Classifier (Test 5, Post-Hoc)

Results. The classifier finds 31.8% of A+ sites carry a founding keyword versus 28.7% in the full corpus (Fisher OR = 1.19, $p = 0.225$ ns). The A++ sub-population (56 sites) shows a stronger signal: 42.9% carry a founding keyword (OR = 1.94, $p = 0.014$ *), consistent with the most precisely aligned sites being disproportionately historically foundational monuments.

The classifier assigns sites to five founding-related categories; full keyword definitions and the site-by-site listing are in the supplementary archive (Tenenbaum, 2026).

References

Aveni, A. F. (2001). *Skywatchers: A Revised and Updated Version of Skywatchers of Ancient Mexico*. University of Texas Press.

Aveni, A. F. (2003). Archaeoastronomy in the ancient Americas. *Journal of Archaeological Research*, 11(2), 149–191.

Casson, L. (1989). *The Periplus Maris Erythraei: Text with Introduction, Translation, and Commentary*. Princeton University Press.

Ciolek, T. M. (2004). *Old World Trade Routes (OWTRAD) Project*. Asia Pacific Research Online, Canberra. http://www.ciolek.com/OWTRAD/DATA/tmcOWTRAD.html

Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.

Crown, A. D. (1989). *The Samaritans*. J. C. B. Mohr (Paul Siebeck).

Freeman, P. R. (1976). A Bayesian analysis of the megalithic yard. *Journal of the Royal Statistical Society, Series A*, 139(1), 20–35.

Good, P. I. (2005). *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer.

Hansen, V. (2012). *The Silk Road: A New History*. Oxford University Press.

Harvey, P. D. A. (1987). Medieval maps of the world. In J. B. Harley & D. Woodward (Eds.), *The History of Cartography, Volume 1: Cartography in Prehistoric, Ancient, and Medieval Europe and the Mediterranean* (pp. 283–370). University of Chicago Press.

Burton, R. F. (1893). *Personal Narrative of a Pilgrimage to Al-Madinah & Meccah*, vol. 2 (Memorial Edition), p. 63, n. 2. Tylston and Edwards.

Hunger, H. & Steele, J. (2019). *The Babylonian Astronomical Compendium MUL.APIN*. Routledge.

Thureau-Dangin, F. (1921). *Numération et métrologie sumériennes*. Revue d'Assyriologie et d'archéologie orientale, 18, 123–142.

Kendall, D. G. (1974). Hunting quanta. *Philosophical Transactions of the Royal Society A*, 276(1257), 231–266.

MacDonald, J. (1964). *The Theology of the Samaritans*. SCM Press.

Magli, G. (2013). *Architecture, Astronomy and Sacred Landscape in Ancient Egypt*. Cambridge University Press.

Magli, G. (2016). *Archaeoastronomy: Introduction to the Science of Stars and Stones*. Springer.

Miksic, J. N. (1990). *Borobudur: Golden Tales of the Buddhas*. Periplus Editions.

Neugebauer, O. (1975). *A History of Ancient Mathematical Astronomy*, 3 vols. Springer.

Powell, M. A. (1987). Masse und Gewichte. In D. O. Edzard (Ed.), *Reallexikon der Assyriologie und Vorderasiatischen Archäologie*, vol. 7, pp. 457–517. Walter de Gruyter.

Purvis, J. D. (1968). *The Samaritan Pentateuch and the Origin of the Samaritan Sect*. Harvard University Press.

Robson, E. (2008). *Mathematics in Ancient Iraq: A Social History*. Princeton University Press.

Ruggles, C. L. N. (1999). *Astronomy in Prehistoric Britain and Ireland*. Yale University Press.

Ruggles, C. L. N. & Cotte, M. (Eds.) (2010). *Heritage Sites of Astronomy and Archaeoastronomy in the Context of the UNESCO World Heritage Convention: A Thematic Study*. ICOMOS & International Astronomical Union.

Ruggles, C. L. N. (Ed.) (2015). *Handbook of Archaeoastronomy and Ethnoastronomy*. Springer.

Sen, T. (2014). The spread of Buddhism. In B. Z. Kedar & M. E. Wiesner-Hanks (Eds.), *The Cambridge World History, Vol. 5: Expanding Webs of Exchange and Conflict, 500 CE–1500 CE* (pp. 447–479). Cambridge University Press.

Silva, F. (2020). A probabilistic framework and significance test for the analysis of structural orientations in skyscape archaeology. *Journal of Archaeological Science*, 118, 105138.

Tenenbaum, S. (2026). *Longitude Quantization in the UNESCO World Heritage Corpus: Domes, Stupas, and 3° Harmonics — Code and Data* (v1.6.0). Zenodo. https://doi.org/10.5281/zenodo.19938283

Tomber, R. (2008). *Indo-Roman Trade: From Pots to Pepper*. Duckworth.

Thom, A. (1964). The larger units of length of megalithic man. *Journal of the Royal Statistical Society, Series A*, 127(4), 527–533.

Thom, A. (1967). *Megalithic Sites in Britain*. Oxford University Press.

Thom, A. (1971). *Megalithic Lunar Observatories*. Oxford University Press.

UNESCO World Heritage Centre. (2024). *World Heritage List* [XML dataset]. Retrieved from https://whc.unesco.org/en/list/xml/.

Watson, R. T. & Horowitz, W. (2011). *Writing Science Before the Greeks: A Naturalistic Analysis of the Babylonian Astronomical Treatise MUL.APIN*. Brill.

Williamson, T. & Bellamy, L. (1983). *Ley Lines in Question*. World's Work.

Data Availability Statement

All data and code are openly available; a permanent snapshot (v1.6.0) is deposited at Zenodo (Tenenbaum 2026; https://doi.org/10.5281/zenodo.19938283); UNESCO XML: UNESCO World Heritage Centre 2024. Scripts, audit files, and maps are indexed in Supplementary S1 (`supplementary/Supplementary_S1.md`), with each file's role, macros produced, and paper section noted. Full reproduction: `reproduce_all_macros.sh`, `update_all_audits.sh`, `generate_figures.py`.

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Conflict of Interest

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Software and AI Disclosure

All statistical analyses were performed in Python 3 using NumPy, SciPy, and pandas. Large-language-model tools were used during preparation of this manuscript: Claude (Anthropic) assisted with prose drafting and editing; GitHub Copilot (Microsoft) assisted with code autocompletion. These tools were used solely as writing and coding aids. All scientific hypotheses, analytical choices, keyword-list decisions, interpretation of results, and conclusions are solely the responsibility of the author. No AI tool was used to generate, fabricate, or substitute for primary data or statistical analysis.